

TABLES

→ Tables are a special Excel feature that converts a data range into a structured dynamic dataset

shortcut: $ctrl + T$

why use Tables?

- automatic filtering
- automatic sorting
- dynamic ranges
- structured references
- automatic formula filling
- better dashboard support

creating a Table

1. select data range
2. press $ctrl + T$
3. check "My table has headers"
4. click OK

Table Design Tab

→ appears when a table is selected

options include:

- Table Name
- Table styles
- header row

- Total row
- Banded rows
- first column

- last column
- remove duplicates
- insert slicer

Dynamic Tables

- when new rows/columns added below table, table expands automatically
- No need to update formulas manually.

Table Name Reference

Ex: $=Jobs$

- returns the entire table as a dynamic array
- can create a duplicate copy of table.
- Column headers are not returned by default.

Return one column

Syntax: $=Table Name [column Name]$

Ex: $=Jobs [job-title-short]$

- returns only that column.

Return Column Header

Syntax: =TableName [[#Headers], [ColumnName]]

Ex: =Jobs [[#Headers], [Salary-year-avg]]

Returns: Salary-year-avg

Return entire column including header

Syntax: =TableName [[#All], [ColumnName]]

Ex: =Jobs [[#All], [Salary-year-avg]]

Returns: header + column Data

Total Row

→ can be enabled from Table Design → Total Row

→ adds a summary row at the bottom

Options:

- sum
- min
- average
- stddev
- count
- variance
- Max

SUBTOTAL()

→ used to calculate summaries while respecting filters

→ Syntax: =SUBTOTAL (function-num, range)

Ex: =SUBTOTAL (9, A2)

Result: Sum of visible rows only

why SUBTOTAL is useful

Suppose: 1000 rows

filter: only data analyst jobs

normal sum: Count all rows

Subtotal: Count only visible rows

AGGREGATE()

→ advanced version of SUBTOTAL

→ Supports

- MEDIAN
- PERCENTILE and many more
- LARGE

Syntax: =AGGREGATE (function-num, options, array)

Table Limitations

1. Cannot contain merged cells
2. Cannot contain completely blank header names
3. Some advanced array formulas may be harder to read.
4. Very large tables may slightly reduce performance

Slicers

→ Provide visual filters for Tables

Path: Table design → Insert slicer

Why use Slicers?

→ Instead of opening filter dropdowns:

click buttons directly

ex: Department

- IT
- HR
- Sales

click ~~buttons~~ directly IT

ex: table instantly shows only IT records

Multi-select in slicers

Method 1 :- Hold ctrl and click multiple values.

ex: IT, HR

Both selected

Method 2 :- Use Multi-select button in top-right of slicer

then click multiple items

No ctrl needed

Tables + Slicers

→ Very common for dashboards

ex: Histogram Dashboard

Slicer: Data Analyst
Data Scientist
Data Engineer

Selecting a job role:



Table updates



Charts update



Dashboard updates

automatically.